

Challenges in recovery and recycling of refrigerants from Indian refrigeration and air-conditioning service sector

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Abstract

India is a large producer and user of chlorofluorocarbons (CFCs) in the refrigeration and air-conditioning (RAC) sector. Government of India has taken several steps to restrict the production and consumption of CFCs. Refrigerant conservation through recovery, recycling (R&R) and reclamation is one way of reducing emissions and encouraging timely phase out of CFCs in developing countries. CFC recovery, recycling and reclamation have been mandated in many developed countries. However, this practice is yet to make an impact in India although it is practiced in MAC sector to some extent. India is planning for the final phase out of CFCs in the RAC service sector, in which R&R will be one of the components.

A model has been developed to assess the economics of R&R for some typical parameters in developing countries like India. The model suggests that the enterprises recycling 1500 kg/a will break even within 1 year for all scenarios. However, R&R may not be cost effective for small workshops and low volume refrigerant vendors until either the price of CFC goes up or the cost of R&R unit is subsidized. A nationwide survey on RAC service sector revealed that in India, there are very few enterprises handling more than 500 kg/a. Therefore, there is a need to provide the RAC service sector with adequate and innovative financial incentives.

This paper attempts to study the issues related to R&R for various sub-sectors of RAC in developing countries with an emphasis on cost effectiveness. India is used as a model for this study.

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1. Introduction

India signed the Montreal Protocol (MP) in 1992. Since then, Government of India has taken numerous response measures to phase out ozone depleting substances (ODS). India was self-sufficient in the production of chlorofluorocarbons (CFCs) and mainly produced and used seven of the 20 substances controlled under MP. With the availability of CFC-11 and CFC-12

from indigenous production, the growth of industries consuming CFCs increased very rapidly. These industries are namely refrigeration and air-conditioning (RAC), solvent, foam blowing, aerosol, etc. RAC sector being a large consumer of CFC-12, refrigerant conservation through recovery, recycling and reclamation is perceived as a way of reducing emissions and encouraging timely phase out of CFCs.

Table 1 presents the ozone depletion potential (ODP) and global warming potential (GWP) of some refrigerants.

From Table 1 it can be seen that CFC-12, the most commonly used refrigerant in India has relatively high ODP and GWP. Generally, RAC sector is the largest consumer of CFC-12. A fairly large amount is consumed

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Table 1
ODP and GWP of some refrigerants

Species	Chemical formula	ODP	100 years GWP
CFC-11	CCl_3F	1	4000
CFC-12	CCl_2F_2	1	8500
HCFC-22	CHClF_2	0.055	1700
HFC-134a	CH_2FCF_3	0	1300
HC-600a	$\text{CH}(\text{CH}_3)_2\text{CH}_3$	0	3
HC-290	$\text{CH}_3\text{CH}_2\text{CH}_3$	0	3

in the service and maintenance of the equipment used in the RAC sector.

Government of India has taken several steps to restrict the production and consumption of CFCs (Ministry of Environment and Forests (MOEF), 1999a, b). The Indian Government has notified the ODS (Regulations and Control) Rules (Ministry of Environment and Forests (MOEF), 2000). The rules prescribe the production sector phase out schedules as well as restrict the expansion of CFC-based products. According to these rules, CFCs in RAC units, manufactured beyond 1 January 2003, have been phased out. The hydrofluorocarbons (HFC) and hydrocarbons (HC) are the alternatives for the commonly used CFCs in RAC sector.

Now, after having phased out the CFCs in original equipment manufacturers (OEM), the target is to phase out CFCs from the service sector. Therefore, the strategy to phase out CFCs from RAC Service Sector (RACSS) has been formulated by Ozone Cell of Ministry of Environment and Forests (MOEF) with assistance from Germany (GTZ), Switzerland (SDC and Infrass), UNEP and UNDP.

Generally RAC units in developing countries are used for an extended period with periodic servicing. Domestic and commercial refrigerators are considered to have life of at least 15 years. Owing to the large population of recently introduced CFC-based appliances that will remain in operation for the next 15 years or so, servicing of these appliances with CFCs may be carried out even after the phase out period, i.e. 2010. Refrigerant conservation through recovery, recycling and reclamation can to some extent reduce the gap between the demand and supply.

According to United Nations Environment Program (UNEP) (1999), recovery means the removal of refrigerant in any condition (vapor, liquid or mixed with other substances) from a system and its storage in an external container. Recycling means reduction of contaminants contained in used refrigerant by separating oil, removing condensables and using devices such as filter dryers to reduce moisture, acidity and particulates. In the case of contaminated refrigerants or mixtures, reclamation or disposal may be necessary.

Reclaimed refrigerant refers to the refrigerant, which has been processed and verified by analysis to meet new product specifications, e.g. in US recovered refrigerant must meet the ARI 700 Standard (Air-Conditioning and Refrigeration Institute (ARI), 1999) (United Nations Environment Program (UNEP), 2000a, b, c).

Destruction means to destroy used refrigerant in an environmentally responsible manner (United Nations Environment Program (UNEP), 2000a, b, c). In some cases different refrigerants get mixed by accident, such mixture can be separated by distillation, but it is expensive. Therefore the easy way is to destroy that mixture. Destruction of CFC is necessary when recycling or reclamation is not cost effective or not possible technically. Destruction plants are in use in developed and in few developing countries. However, this practice does not exist in many developing countries, including India.

The advantage of recycling is that it can be done onsite and the refrigerant can be immediately charged into the equipment. Recycling may be done using a single pass system or a multi pass system. Though recycling gives a considerably pure refrigerant and is cheaper than reclamation, the recovered refrigerant may not match the specification of a virgin product. Therefore, it is imperative to use certified R&R units under such situations. There are no inexpensive instruments to measure the contaminant levels of the refrigerants.

Many international companies are now producing R&R machines on a large scale. Many of them are certified for the performance and quality of recovered refrigerants, as per some standards. The costs of these machines are in the range US\$ 2300–4000. These units have a recovery rate of 0.27–1.36 kg/min.

Reclamation means transportation of recovered refrigerant to a central processing facility and subsequent cleaning by filtering, drying and chemical treatment to meet the expected purity standards. Reclamation, certainly a better process, is ideally suited for large volume of refrigerant. Its main disadvantage is that it cannot be applied for small systems and the infrastructure required is elaborate and expensive. Many international companies are in this business. However, there is no reclamation facility in India, so far.

2. International experiences

R&R is being practiced in many developed and developing countries. In many developed countries, there are regulations in place to prohibit the venting of ODSs and thus making R&R mandatory. A short review of the existing regulations of some developed countries is presented below.

2.1. Regulations

The USEPA—Federal Register 40 CFR part 82, 1996 and the ARI standard 700 (*Air-Conditioning and Refrigeration Institute (ARI), 1999*) specification for fluorocarbon refrigerant have set the guidelines for R&R practices. These regulations have laid down rules like prohibition on venting, service practice, and equipment certification and leak tightness of systems, technician certification, sales restrictions, safe disposal and record keeping requirements. Inability to follow these regulations means a heavy penalty.

In the European Council (EC), regulation no. 3093/94 requires that all precautionary measures practicable shall be taken to prevent leakage of CFCs and HCFCs. It also prescribes the minimum qualification requirements for servicing personnel. In future, an annual leak tightness inspection could be made mandatory for installation containing CFCs and HCFCs.

In the French Decree no. 971271 of December 1992, the French government made it mandatory to recover CFCs, HCFCs and HFCs for any installation containing more than 2 kg charge. They also make it mandatory to conduct annual leak tightness inspections for all sectors except for domestic and automobile air conditioning sectors.

The Netherlands, in its decree, describes the conditions for leak tightness of systems and also the frequency of leak tightness inspection, e.g. once a year for charges under 3 kg, once every 3 months for more than 30 kg, once a month for more than 300 kg. It also specifies the requirements of leakage detection and monitoring systems.

The United Kingdom Environmental Protection Act of 1990 prohibits the venting of refrigerants during servicing/decommissioning, adding refrigerant to a leaking system. It also mandates to use a vacuum pump to evacuate and to limit emissions.

2.2. Achievements and experience gained by developed countries

Many countries have introduced R&R programs. Some are described below.

In France, a gradual evolution in the efficiency of the recovery program has been achieved. Initially, the quantity recovered and reclaimed was low but after the implementation of the recovery rule and carrying out deposit-refund schemes, the quantity increased noticeably.

In Denmark, the recycling of CFC refrigerants was started in late 1970s, immediately after the recycling machines were developed. After the ban on imports and sale of CFCs from 1 January 1995 by the Danish Government, the market prices for CFCs increased as high as US \$ 50/kg. This promoted the use of R&R

significantly. The refrigerators are collected and transported to a central recycling center where the refrigerant is recovered and reclaimed or destroyed as per the quality of the refrigerant. A surcharge is levied while selling a new refrigerator to cover the cost of the disposal of old refrigerators (*United Nations Environment Program (UNEP), 1999*).

In the United States, some major companies producing R&R units are playing a positive role in encouraging R&R. These companies provide recovery, recycling or reclamation services customized to the industry's needs. They employ programs like exchanging of used refrigerants for non-CFC refrigerants, exchanging excess of refrigerants for cash, cylinder pick-up and delivery, disposal of refrigerant, refrigerant separation and even oil decontamination or disposal. They also provide empty recovery cylinders and arrange for laboratory testing of refrigerants as per ARI standard 700 (*Air-Conditioning and Refrigeration Institute (ARI), 1999*).

In European Union, a number of countries have included in their R&R schemes, HCFCs and HFCs too, along with CFCs. In some countries such as Austria, parts of Germany and Denmark, the recovery from domestic refrigeration equipment is financed by a deposit or fee. In the absence of regular reporting or monitoring of R&R scheme, any assessment of the amounts of ODS recovered, recycled and destroyed is difficult (www.gtz.de).

In developed nations, due to regulations and hence the high prices of CFCs, e.g. indirect tax or ban on the import, the recycling of CFCs is considered to be a profitable. In these nations, the attitude regarding CFC recycling is positive among the refrigerant service companies, the public and owners of refrigeration facilities.

2.3. Achievements and experience gained by Countries with Economy in Transition

There is a mixed experience with R&R schemes in the Countries with Economies in Transition (CEIT). In Czech Republic, refrigerant was recovered but not reclaimed because the market price of the recycled refrigerant was found higher as compared to the virgin material (*Global Environment Facility (GEF), 1999*). This experience suggests that R&R cannot be steered through prices alone. Price incentives have to complement suitable regulations. Import fees might be a means to control the market price so as to create incentives for recovering and recycling. This experience of CEITs is particularly relevant for the implementation of R&R schemes in developing countries. Because of the varying conditions in countries, there is probably not a single design of the scheme. Rather, a flexible approach may be required which would allow adaptations to local

circumstances as well as new developments such as changing prices (www.gtz.de).

2.4. Achievements and experience gained by developing countries

Refrigerant Management Plans (RMP) are being implemented in many developing countries consuming low volumes (LVC) of ODS in critical refrigeration sectors. The objective of the RMP is to design and implement an overall integrated strategy for cost effective phase out in a coordinated and planned manner. This could be achieved with adequate training, containment, retrofitting, introducing new technology and conducting R&R programs.

UNIDO has reported lessons learnt mostly from six projects in African countries (United Nations Environment Program (UNEP), 2000a, b, c). According to it, the implementation of R&R scheme in LVCs is a very efficient means to reduce the emission of CFCs. However, specific training programs must absolutely support the supply of equipment for trainers and for technicians to implement good refrigeration management practices.

The practice of using R&R equipment is not well established in the Philippines as the price of CFC-12 has been relatively low and the supporting infrastructure (e.g. reusable cylinders) is currently only available to a limited number of service companies. In the absence of reclamation facility, enterprises that have recovered CFC report problems in identifying where to send the recaptured refrigerant for reclaiming (www.sei.se).

In a developing country like Ghana, the National Ozone Unit (NOU) focus on training workshop managers and service technicians to minimize unnecessary ODS emissions during servicing and maintenance of refrigeration equipment through good servicing practices, including R&R. The NOU designed and implemented a complementary program, which included the establishment of R&R scheme. One of the key elements of the project was establishment of subsidy for the purchase of R&R equipment (www.uneptie.org).

The review of international scenario suggests that there is a lack of information and awareness about R&R, drop-in substitutes and alternative refrigerants is the primary concern. Both technicians and consumers need to be convinced about their role to heal the ozone layer, as only passing regulations to control emissions and thereby enforcing R&R may be difficult to implement. Further, service technicians need to be equipped and trained. Financial incentives encourage R&R to some extent as expensiveness of the R&R systems is acknowledged the major hurdle. With the R & R equipment on hire, instead of owning, the investment on equipment can be reduced. Then the operating costs will be only the cost for transport and labor.

3. Standards for purity of recycled refrigerant

Standards have a great influence because they are widely used as a basis of legislation and regulation at all levels of government, from local to international. Standards are often referenced by government agencies and in designs and specifications (United Nations Environment Program (UNEP), 2001).

International Standards Organization (ISO), Air-conditioning and Refrigeration Institute (ARI) and United States Environmental Protection Agency (USEPA) are some of the agencies setting standards or providing the guidelines for R&R. These are generally followed by developed countries. For developing countries, the national standards agencies may provide the suitable guidelines mostly following the international standards with some dilution, e.g. Bureau of Indian Standards (BIS).

High-purity refrigerants are required for air-conditioning and refrigeration. When equipment needs servicing, technicians use recovery machines to ensure that the refrigerant is not vented to the atmosphere. ARI has developed standards, e.g. ARI 700 (Air-Conditioning and Refrigeration Institute (ARI), 1999) and ARI 740 (Air-Conditioning and Refrigeration Institute (ARI), 1998) to assist the industry in evaluating the quality of refrigerants and the performance of refrigerant R&R equipment (www.ari.org).

The purpose of ARI 700 standard (Air-Conditioning and Refrigeration Institute (ARI), 1999) is to establish purity specification and specify the associated methods of testing for acceptability of fluorocarbon refrigerants regardless of source (new, reclaimed and/or repackaged) for use in new and existing refrigeration and air conditioning products within the scope of ARI. Many developed countries have standards for refrigerants. Now they are formulating guidelines for recycled refrigerants.

These standards are intended for the guidance of the industry including manufacturers, refrigerant reclaimers, repackagers, distributors, installers, servicemen, contractors and users. In India, there is a similar standard.

The Indian Standard (IS, 1993) prescribes requirements and methods of sampling and test for chloro-fluorohydrocarbons for RAC purposes and for use as solvents and for making aerosol. However, there is no such standard for R&R in India yet.

In US, standards are in place to provide more flexibility where refrigerants are transferred between appliances with different ownership. They address third party certification, record keeping, technician certification and sales restriction (Environmental Protection Agency (EPA), 1996).

It is not always practicable or feasible in a service workshop, to confirm that a recycled refrigerant meets standard level by test. Therefore guidelines are written

to give the servicing contractor some criteria to help to determine the options whether the recovered refrigerant can be recycled or not (IRG-2, 1994).

Table 2 presents various standards for maximum allowable contaminant levels in refrigerant CFC-12. IS: 5610 is for virgin refrigerant, while EPA and Industry Recycling Guide 2 (IRG-2) are for recycled refrigerant and ARI 700 (Air-Conditioning and Refrigeration Institute (ARI), 1999) is standard for specifications of refrigerant regardless of source (new, reclaimed and/or repackaged).

4. Overview of Indian RACSS

The service sector in India is made up of a diverse range of establishments, which provide maintenance services to all types of RAC systems. The sector consists of both formal and informal enterprises. The informal sector consists of small and micro-enterprises (SMEs), which might in many cases, refer to only one or two persons who would be full time involved and most of the technicians might have learned about servicing in the field rather than through formal training.

Despite its limitations, the nationwide survey revealed that the total consumption in the Indian RACSS was about 1200–1300 MT in 2000. This survey did not include consumption at institutional owners with own service personnel such as Indian Railways (Batra, 2002). These findings were close to the findings of Devotta and Asthana (2000). The estimated demand is expected to be about 700 MT in 2010, out of which 30% will be consumed for the servicing of domestic refrigerators, 33% for commercial appliances and the rest 37% for MAC (Devotta and Asthana, 2002). Production of CFCs in India by 2010 will be phased out leading to a wide gap between the demand and the supply. It is

probable that total destruction will be fairly low in refrigeration industry, since demand for CFC will remain high.

The study of various sub-sectors of the Indian RAC sector (Devotta and Asthana, 2000) revealed that R&R is not yet practiced in domestic and commercial sub-sectors. However, there is growing awareness regarding it through the ongoing Human and Institutional Development in Ecological Refrigeration (HIDECOR) project. As the CFC-12 charge quantity in these refrigerators (majority of 165–200 l) is very small, and the residual CFC-12 in system is likely to be still much smaller, R&R may not be a very cost effective option in this sector. This merits special considerations, if R&R has to be encouraged in this sub-sector. Large commercial and industrial units, e.g. refrigerated cabinets and ice candy machines have some potential for R&R. The growth rate for this sector is high and the charge quantity in these units is reasonably high.

Mobile air-conditioning (MAC) accounts for a considerable share of the total ODS consumption. There is a frequent recharging of the appliance during its lifetime due to substandard road conditions and the hot and humid climate that promotes corrosion. The high growth in the automobile sector and the increasing use of car air-conditioners is likely keep this sector as the main target sub-sector, with a high potential for R&R.

There is not much R&R of CFC-12, as the servicing of CFC-12 based MACs is generally provided by unorganized sector. Devotta and Asthana (2000) revealed that “Topping-up” is a common practice in servicing of MACs. However, MNC car and MAC manufacturers, using HFC-134a, are already practicing R&R at their service centers, even though it is not mandatory and may not be economically viable. This is purely based on global corporate policies. MNCs service their units in their service centers, which are fairly

Table 2
Standards for maximum allowable contaminant levels in CFC-12

Parameter	Reporting units	EPA	ARI 700	IRG-2	IS: 5610
Boiling point	°C at 101.3 kPa	—	–29.8	—	–29.8
Boiling point range	°C	—	0.3	—	0.3
Vapor phase contaminants					
Air and other noncondensables	% by volume	2.0	1.5	2.0	1.5
Liquid phase contaminants					
Water	ppm by weight	10	10	10	10
Volatile impurities including other refrigerants	% by weight	2.0	0.5	2.0	—
High boiling residue	% by volume	0.02	0.01	0.02	—
Particulates or solids	Visually clean to pass	Visually clean	pass	Visually clean	—
Acidity	ppm by weight (as HCL)	1.0	1.0	1.0	pass
Chloride	No visible turbidity	No turbidity	Pass	No turbidity	—

equipped, and the car owners, elite consumers, do not mind paying extra as they have no other choice. This sector should be a key sector for any R&R demonstration project.

5. Economic model for R&R

In order to assess the economics of the R&R, a model has been developed taking into consideration some of the parameters applicable to Indian RAC servicing sector (Devotta and Joshi, 2002). The key features/assumptions of the economic model are as follows:

1. Life of the R&R unit is 10 years.
2. Efficiency of the R&R unit is 90% (The efficiency of an R&R unit is defined as the ratio of the refrigerant recovered to the refrigerant charge in the refrigeration unit).
3. Recycling rate of the typical R&R unit is 27 kg/h (0.45 kg/min).
4. Various economic parameters are summarized in the Table 3.

The payback period (PBP) is defined as (Holland et al., 1983):

$$(PBP) = C_{FC} / (A_S - A_{WC}), \quad (1)$$

where C_{FC} is the cost of the R&R unit & accessories, A_{WC} is the annualized working cost of the unit & accessories, A_S is the recovered cost of recycled refrigerant per annum, $A_S = M_{CFC} c_{CFC} M_{CFC}$ is the recycling rate (kg/h) $\times$ annual hours of operation.

The annualized working cost of the unit is given by the following equation:

$$A_{WC} = A_{FC} + A_{OP},$$

where A_{FC} is the annualized fixed capital cost of the unit and accessories, A_{OP} is the annualized operating cost of the unit and accessories, including cost of labor, electricity and maintenance/spares.

Annualized fixed capital cost of the unit A_{FC} is given as:

$$A_{FC} = C_{FC} f_{AP},$$

$$f_{AP} = i(i+1)^n / [(i+1)^n - 1],$$

Table 3
Economic parameters in the model for R&R

Variable	Range	Typical value
Cost of R&R unit (INR ^a million)	0.15–0.45	0.3
Unit cost of CFC (INR/kg)	300–600	300
Interest rate (%)	5–12	12
CFC-12 Recovered (kg/year)	250–1500	250

^a 1 US\$ is about 50 INR.

where f_{AP} is the annuity present worth factor, i is the annual interest rate, n is the life of the unit in years.

5.1. Results and discussion

Fig. 1 presents PBPs for an enterprise recycling 250 kg/a of CFCs. A nationwide survey (Batra, 2002) suggests that there is a large population of enterprises handling 100–200 kg/a and as compared to the enterprises handling 200–500 kg/a (Batra, 2002). The economic model reveals that break even in less than 3 years is possible only if the cost of R&R unit is INR 0.15 million and the cost of CFC-12 is INR 300/kg. Therefore, it is not economically attractive for small workshops and low volume refrigerant vendors until the price of CFC goes up or R&R unit is subsidized.

Fig. 2 presents PBP for case for enterprises recycling 750 kg/a. The economic model suggests that the break even is less than 3 years for all possible scenarios for them. If the cost of R&R unit is less than INR 0.3 millions, then break even is possible even in 2 years. Therefore, it is very attractive for medium volume vendors. It may be worth noting that in India, there are very few enterprises handling more than 500 kg/a.

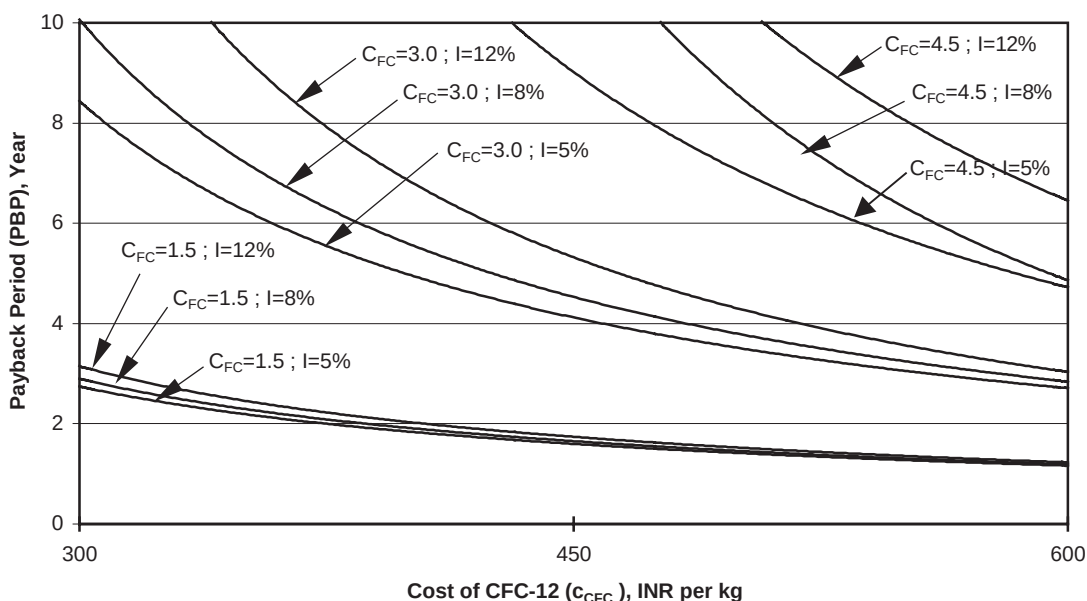
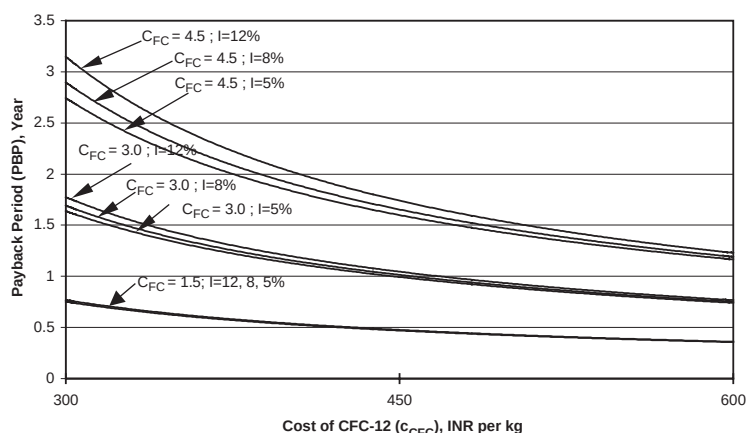
Fig. 3 presents the case for an enterprise recycling 1500 kg/a. Break even within 1 year is possible for all scenarios for this category. However, there are a negligible number of enterprises falling under this range.

5.2. Barriers for R&R in developing countries

Making a significant reduction in atmospheric concentrations of greenhouse gases does require international coordination. However, the economic incentive for any nation, particular for developing countries, to reduce emissions unilaterally is marginal. To be cost-effective, international climate change policies should include subsidies and incentives, in the initial stages, as a complement to MP. The current limits for the unit abatement cost for CFC-12 in RAC sector are grossly inadequate for cases of fragmented and low CFC-12 consuming service sector.

The international carbon offset market in 2002 was about US\$ 2.5/ton of CO₂ equivalent (Karani, 2002). Market in some developing and CEIT regions is about US\$ 10–15/ton of CO₂ equivalent. It is expected to rise up to US\$ 30–40/ton of CO₂ equivalent in those regions. This suggests that the current unit abatement cost is too low especially for abatement from the service sector, which is fragmented and dispersed with very low CFC consumption.

A nationwide survey organized by the Government of India under the RAC service sector phase out plan (Batra, 2002) has revealed that most of the SMEs in India have an annual turnover less than INR 0.5 million. The high cost of a quality R&R unit (approximately

Fig. 1. PBP vs. Cost of CFC-12 for $M_{\text{CFC}} = 250$ kg/year.Fig. 2. PBP vs. Cost of CFC-12 for $M_{\text{CFC}} = 750$ kg/year.

INR 0.15 to 0.3 million) is beyond their financial reach. The idea of a recovery and reclamation network may not go well with Indian RAC service sector because, the volume of CFC recovered would be too low and the process would be time consuming and also involve transportation cost. Therefore, adequate and innovative financial incentives have to be created for the participation of SMEs in R&R.

Currently IS 5610 (1993) is the mandatory standard for virgin CFCs being sold in India. If the quality of the reclaimed refrigerant has to be assured, then contaminated CFCs have to be reclaimed using certified R&R machines, preferably in organized facilities.

Although there are some international standards, e.g. ARI, SAE, ISO, there is a need to have country specific standards for R&R. These standards should be practical for local implementation with some prescribed guidelines for the industry. Therefore, developing countries should define standards for the performance of R&R, maximum allowable contaminants in the recycled refrigerants. Procedure to ensure the quality of the recycled refrigerants is an important issue in R&R scheme. This should be featured strongly in training courses for users of recycling machines. Further, there is a need to establish a well-equipped national-level test center.

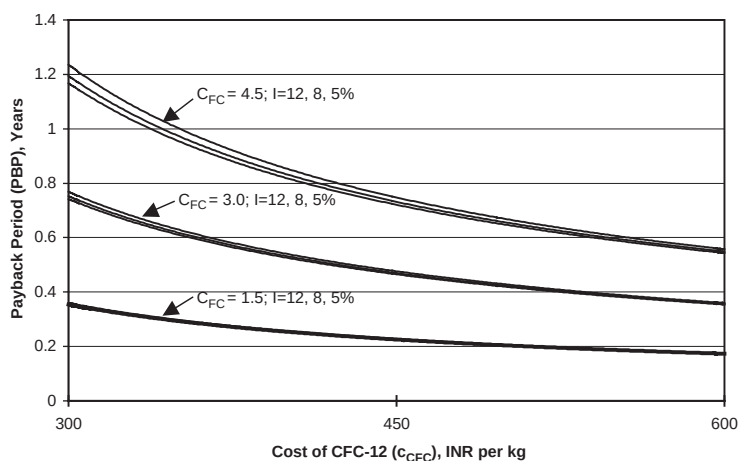


Fig. 3. PBP vs. Cost of CFC-12 for $M_{CFC} = 1500$ kg/year.

Currently the cost of CFC-12 in India is about INR 400/kg, many fold the international price, in the range US\$ 1.50 to 3.0. R&R of CFCs is not practiced mainly because of the easy availability of CFCs in India. The current cost of HFC in India is about INR/400 kg compared to the international price of US\$ 3–4. Still, HFC-134a is recycled to some extent in MAC sector mostly promoted by MNCs. The projections of CFC consumption suggest that there may be a gap between demand and supply only after 2007. If R&R has to be introduced immediately, then some innovative and incentive mechanisms have to be introduced.

5.3. Strategies for R&R in India

There are many challenges for the success of R&R in India, as it may not be economically driven. There is only a small margin between conditions, which are economically attractive and unattractive. This margin is where institutions like Multilateral Fund under the MP or Global Environmental Facility should support SMEs in developing countries. Such a support can provide the incremental costs to make R&R a profitable project. The role of training is important as it raises the awareness on the environmental benefits of R&R. A comprehensive training program to cover the entire country to promote R&R should be launched.

There is a large population of SMEs in India having very low CFCs consumption. For them R&R is not economically feasible. Under these circumstances, decentralized vs. centralized R&R systems needs to be reviewed.

In decentralized R&R systems (United Nations Environment Program (UNEP), 1999), the participating service workshops are equipped with both recovery and recycling units. R&R is either performed locally at the customer's premises, using mobile R&R equipment, or

at the service workshop. The recycled refrigerant is normally used to recharge the same refrigeration unit from which it was recovered. The recycling process is controlled by the local workshop. This is suitable for enterprises whose CFCs consumption is high.

In centralized R&R systems (United Nations Environment Program (UNEP), 1999), the participating service workshops are equipped with recovery machines only, and the recovered refrigerant is recycled in a recycling center. In some cases, the recycling center can also be equipped with a reclaim station. The recycling center controls the recycling process. This system will be suitable for small enterprises mushroomed in a particular geographical location.

Indigenous development of R&R systems should be encouraged to keep the initial cost down. Imported R&R units are fairly expensive. In case of import, R&R units should be procured with a long-term supply of spare parts. In many cases without support service, units imported are lying unused for want of spares.

Further, there is a need to evolve financial mechanisms to support R&R. This can be done by considering innovative financial incentive packages, e.g. deposit-refund for CFCs, R&R unit hiring scheme, low interest loans, import duty exemption, etc.

Experience in the US suggests that standards for recycled refrigerants should be specified keeping in view not only OEMs but also the service sector. In the US, Regulations Implementing Section 608 of the Clean Air Act Amendments of 1990 required that used refrigerants should be reclaimed to purity of ARI standard 700 (Air-Conditioning and Refrigeration Institute (ARI), 1999) prior to sale to a new owner. At that time, EPA established sunset for these rules and asked the public to develop a more flexible set of rules. In response, ARI and a number of other industry associations developed the IRG-2, which provided contractors with greater

flexibility to extend service while retaining ARI 700 (Air-Conditioning and Refrigeration Institute (ARI), 1999) purity levels for refrigerant samples. However, EPA was unable to adopt IRG-2 to take the place of the original reclaim rules, and on 27 December 1996, the Agency extended the existing regulations until such time as it could develop an alternative approach.

If the service sector in an advanced country like US has problem in accepting the similar purity of virgin CFCs, then it is unlikely that such a stringent standard will ever work in a developing country like India. One has to be pragmatic about the environment benefit of R&R and the cost associated with it. If one were to make an assessment of percent of refrigerant recycled of the total CFCs used in a year, it appears that this percentage is rather low even in developed countries. Therefore, most of the damage to the environment is caused by the CFCs not recycled. This low rate of recycle may be attributed to the stringent quality of refrigerant prescribed. However, one cannot ignore the need for some quality of recycled refrigerants and the R&R unit to ensure the efficiency of recovery.

For an effective implementation, certification programmes for R&R systems and operators and continuous monitoring and review of R&R programme is required. For performance testing of the R&R units developed indigenously, testing standard for recovery/recycling units, similar to those specified in ARI-740/98 (Air-Conditioning and Refrigeration Institute (ARI), 1998) needs to be established. The specifications of methods of testing for rating, performance test procedures, sampling and chemical analysis methods are necessary.

The gas cylinder rules in India need to be reviewed and revised to facilitate servicing (small cans) and recovery cylinders. The current rules do not permit filling of cylinders while R&R would entail filling of recovered refrigerants into temporary cylinders for transportation. There is a need to assess the existing rules and revising them to facilitate R&R, instead of being an impediment. Without such positive initiatives, R&R may not be fully realized in India.

6. Conclusions

Developing nations in their RMPs, must formulate a national policy for R&R reclamation and destruction considering the current international trends, including the success and failures.

A comprehensive and innovative financial incentive packages should be worked out to promote R&R in developing countries. This will be a critical factor considering that CFCs will be available till 2010 as per the existing CFC production phase out. Multilateral fund supports for such schemes should be considered.

The program should be continuously monitored, reviewed and appropriate corrective measures should be taken.

Direct regulations are less flexible than financial incentives and are more difficult to enforce in developing countries like India. However, direct regulations may be brought about for certification programs for equipment and operators and for containment standards through training institutes like Industrial Training Institutes (ITI).

There is a need to review existing standards and rules and to consider appropriate revisions to encourage R&R. Studies on emission sources must be undertaken to arrive at the methodologies/technologies to mitigate the leakage problems. Better standards for design, manufacture, safety and servicing should be evolved.

There is a need to create awareness on R&R for both industry and consumers. It may be certainly economical option for large commercial and MAC sectors that needs to be demonstrated. Like in India, a survey should be carried out to identify large CFC consuming regions and focus should be given on these regions in demonstrating and establishing R&R. Further, measures should be taken to stop smuggled CFCs in the country.

The supply of R&R equipment and the establishment of R&R systems alone do not ensure the successful operation of such systems. They must be supported by regulatory structures involving command and control tools as well as incentives for the end-users. The establishment of such R&R systems represents a cost-effective step in reducing the consumption of virgin CFC refrigerants without major capital investment, and in allowing existing CFC-based equipment to run until the end of its economic life.

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